The predictions that the half-lives of some heavy elements may be very long have led to searches for evidence of their existence in nature. Several reports of tentative evidence for such superheavy elements have appeared but subsequent experimentation has explained the observations as based on known elements with only a few exceptions.

14.3. ACTINIDE PROPERTIES

14.3.1. The actinide series

As early as 1923 N. Bohr suggested that there might exist a group of 15 elements at the end of the Periodic Table that would be analogous in their properties to the 15 lanthanide ("rare earth") elements. This idea, combined with the increasing stability of the +3 oxidation state for the transuranium elements as the atomic number increases from $Z = 93$ to 96, led Seaborg to the conclusion that these new elements constituted a second rare earth series whose initial member was actinium. As the atomic number increases from 90, electrons are added in the 5f subshell similar to the occupation of the 4f subshell in the lanthanides, see Table 14.3. This series would be terminated with element 103 since this would correspond to the addition of 14 electrons for a completed 5f subshell.

Table 14.3 Electronic configuration, radii (in picometres, pm), and oxidation states in (acid, non-complexing) aqueous solution

Atomic number	Element	Metallic radius (pm)	Atomic (g)† config.	Effective ionic radius†		Oxidation* states
				M^{3+}	M^{4+}	
89	Ac	188	$5f^0 6d 7s^2$	111.9		**3**
90	Th	180	$5f^0 6d^2 7s^2$	(108)	97.2	(3) **4**
91	Pa	163	$5f^2 6d 7s^2$	(105)	93.	(3) 4 **5**
92	U	156	$5f^3 6d 7s^2$	104.1	91.8	3 4 5 **6**
93	Np	155	$5f^4 6d 7s^2$	101.7	90.3	3 4 **5** 6 (7)
94	Pu	160	$5f^6 7s^2$	99.7	88.7	3 **4** 5 6 (7)
95	Am	174	$5f^7 7s^2$	98.2	87.8	**3** 4 5 6
96	Cm	175	$5f^7 6d 7s^2$	97.0	87.1	**3** 4
97	Bk		$5f^9 7s^2$	94.9	86.0	**3** 4
98	Cf		$5f^{10} 7s^2$	93.4	85.1	(2) **3**
99	Es		$5f^{11} 7s^2$	92.5		(2) **3**
100	Fm		$5f^{12} 7s^2$			(2) **3**
101	Md		$(5f^{13} 7s^2)$	89.6		2 **3**
102	No		$(5f^{14} 7s^2)$			**2** 3
103	Lr		$(5f^{14} 6d 7s^2)$	88.2		**3**

*The most stable oxidation state in (acid, non-complexing) aqueous solution is given in bold, most unstable (or only found in the solid state) within parenthesis.
†Electronic configuration and ionic radii (coordination number 8) according to Katz, Seaborg and Morss and to Seaborg and Loveland.

Seaborg's actinide hypothesis was initially a subject of considerable objection since the trivalent oxidation state, unlike in the lanthanide family, was not the most stable in aqueous solution for the elements between $Z = 90$ and 94. In aqueous solution the most stable oxidation states are +4 for thorium, +5 for protactinium, +6 for uranium, +5 for neptunium, and +4 for plutonium, see Table 14.1, right column. Only for the elements beginning with americium is the +3 the most stable state in solution. Seaborg, however, had correctly identified Ac as the precursor (analogous to La) and Cm as the midpoint element (analogous to Gd). Recent investigations have shown that mendelevium and nobelium have a divalent state in solution (which probably is the most stable for nobelium). This corresponds to the divalent state observed for ytterbium in the lanthanide elements. For $Z = 90$ to $Z = 94$ the 5f and 6d orbitals are very close in energy and the electronic occupation is variable. However, for these elements one should use Schrödinger orbitals and their energies with considerable caution due to the limited velocity of light in our universe, cf. §2.4.

The recognition of the similarity in chemical properties between the actinide and lanthanide elements was an important contributing factor in the synthesis and isolation of the transcurium elements. Most of the chemical identification was carried out by eluting the elements from columns of cation exchange resin. The pattern of the elution behavior from the resin bed of the lanthanide elements made it possible to predict with good accuracy the expected elution position for a new actinide element (Fig. 14.8). This technique constituted the most definitive chemical evidence in the discovery experiments for the elements from atomic numbers 97 through 101. More recently these conclusions have been confirmed by spectroscopy.

Orbital designations names for solutions to the Schrödinger equation will be used here, although their validity is questionable for these heavy elements. We do this because the more correct descriptions based on Dirac wave functions are seldom used in chemistry and their multiple and multidimensional shapes are difficult to visualize.

The electronic ground state of $_{89}$Ac contains four filled inner electronic shells (the main quantum numbers 1, 2, 3 and 4, designated K, L, M and N, and containing 2, 8, 18 and 32 electrons, respectively); see Ch. 6. The subsequent outer 5th, 6th, and 7th shells (designated "O", "P" and "Q", respectively) are partly empty. In the symbolism used to designate the electronic structure of an atom, the filled shells are usually omitted, as they do not contribute to the chemical properties (and interactions) of the atom. The symbolism $5s^2p^6d^{10}f^06s^2p^6d^17s^2$ for $_{89}$Ac indicates that in the 5th shell the subshells s, p, d and f contain 2, 6, 10 and 0 electrons, respectively; the 6th subshell contains 2 s-electrons, 6 p-electrons and 1 d-electrons; the outermost 7th shell contains 2 s-electrons. Remembering that the s-, p-, d- and f-subshells can accommodate 2, 6, 10 and 14 electrons respectively, these four outer shells are partly empty. Also leaving out the filled sub-shells, the electronic structure for Ac can be written $5f^06d^17s^2$; though the 2 7s-electrons could have been left out we keep them for specific purpose. The construction of the electronic

configuration for gaseous atomic actinide atoms in Table 14.3 then becomes obvious. These electronic structures have been determined spectroscopically.

From Table 14.3 it is obvious that there can be interaction between the 5f, 6d and 7s orbitals, as a consequence of their very similar binding energies. As these partly empty outer orbitals are the ones which may take part in the chemical bondings we can expect the actinides to show a highly variable chemistry.

The main feature of the electronic structure, when going from $_{89}$Ac to $_{103}$Lr is the successive filling of the 5f subshell. This leads to a slight contraction of the atomic and ionic radii with increasing atomic number (the *actinide contraction*), as seen from several columns. However, there are irregularities, which are attributed to the extra stability of the half-filled 5f subshell (at 7 electrons), leading to more labile 6d-electrons.

14.3.2. The solid state

The actinide metals can be produced by heating the tri- or tetra-fluoride with metallic alkaline earth or alkali elements:

$$PuF_4 + 2Ca \rightarrow Pu + 2CaF_2 \qquad (14.24)$$

The metals exhibit several allotropic modifications: 3 for uranium and 6 for plutonium between room temperature and the melting point (1130°C for U and 640°C for Pu) (Fig. 14.10). The density of the actinide metals at room temperature shows an unusual variation: Ac ~10 000, Th 11 800; Pa 15 400; U 19 100; Np 20 500; Pu 19 900; Am 13 700; Cm 13 500; Bk 14 800 kg m^{-3}.

All the metals are very electropositive and attacked by water vapor with production of hydrogen. They are slowly oxidized in air and at higher temperatures; in the form of small chips, they are pyrophoric. The hydrides, oxides, nitrides, and halides are produced most easily by heating the metals in the appropriate elemental gas. The fluorides are among the most important solid actinide compounds since they are the starting material for the production of the metals. The volatile hexafluoride of uranium is used in isotopic enrichment (Ch. 3). The preparation and properties of Th and U are described in §§14.2.2.2 , 14.2.4.2, and 14.2.4.3. Actinide chemistry is also discussed in §§21.5 and 22.6.

14.3.3. Actinide oxidation states

The electrons in the 6d and 7s subshells are more loosely bound than the electrons in the filled subshells, and, in general, also than the 5f electrons. In these outer shells the binding energies are in the range of a few eV, i.e. the same order of magnitude as is common in chemical bonding. Thus, from Table 14.3, it is understandable that Ac easily looses its 6d7s^2 electrons to form Ac^{3+}, and Th its 6d^{2}7s^2 electrons to form Th^{4+}. For the subsequent elements, from Pa to Am, the situation is more complicated. There are reasons to

suppose that the spatial characteristics of the f-subshell orbitals may change abruptly at certain atomic numbers; that is, the f-shell electrons may be shielded more strongly in some elements than in others where the f-orbitals extend close to the surface of the electronic cloud (where chemical interaction occurs), and where the 5f-electrons are in closer contact with the d- and s-shell electrons. There can be little doubt that the 5f-electrons are present in all of the actinide elements after Pa.

The chemical properties of the actinide elements have been intensely studied for the elements available in at least microgram quantities (Z 90 - 99) but much less so for the heaviest members of the family (Z 100 - 103). Table 14.3 lists the valency states. In Figure 14.7 redox diagrams are given for the most important actinides; for comparison, standard potentials are included for some useful redox reagents. Any particular actinide can be obtained in a desired valence state by the use of proper oxidizing or reducing agents (Table 14.4).

The pentavalent state of the actinides (except for Pa and Np) is less stable than the other states and normally undergoes disproportionation in acid solutions. Plutonium is particularly interesting in the variety of oxidation states that can coexist in aqueous solutions (Fig. 14.7, at 1000 mV). For example, a plutonium solution in 0.5 M HCl of 3×10^{-4} M Pu concentration at 25°C, which is initially 50% Pu(IV) and 50% Pu(VI), will equilibrate within a few days via disproportionation reactions to an equilibrium system that is 75% Pu(VI), 20% Pu(IV), and few percent each of Pu(V) and Pu(III), in the absence of complexing anions.

The reactions are:

$$2PuO_2^{+} + 4H^{+} \rightleftarrows Pu^{4+} + PuO_2^{2+} + 2H_2O \tag{14.25}$$

and

$$PuO_2^{+} + Pu^{4+} \rightleftarrows PuO_2^{2+} + Pu^{3+} \tag{14.26}$$

Although Pu(V) readily disproportionates at concentrations $\geq 10^{-8}$ M in acidic solutions, it is the state observed in more basic natural and ocean waters, partly due to the stability of the bicarbonate complex, cf. Ch. 22.

14.3.4. Actinide complexes

Since the differences in energy of the electronic levels are similar to chemical bond energies, the most stable oxidation states of the actinides may change from one chemical compound to another, and the solution chemistry will be sensitive to the ligands present. Thus complex formation becomes an important feature of the actinide chemistry (cf. §18.4.3 and §22.7).

The chemical properties are different for the different valency states (Table 14.5), while in the same valency state the actinides closely resemble each other. These

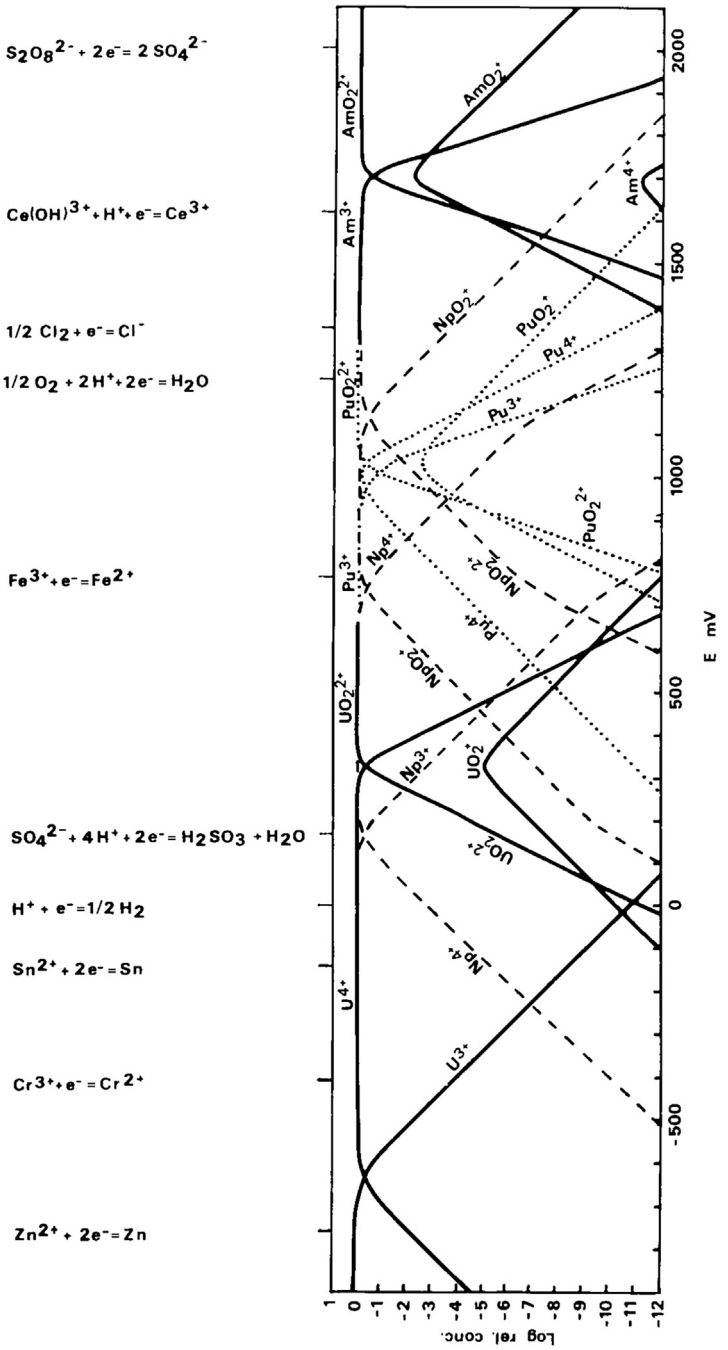


Figure 14.7 Redox diagrams for U, Np, Pu, and Am in 1 M $HClO_4$ at 25 C. Each actinide is assumed to be in equilibrium in its own solution at the indicated redox potential.

Table 14.4 Preparation methods and stability of actinide ions in aqueous solutions

Ion	Stability and method of preparation
U^{3+}	Slow oxidation by water, rapid oxidation by air, to U^{4+}. Prepared by electrolytic reduction (Hg cathode).
Np^{3+}	Stable to water, rapid oxidation by air to Np^{4+}. Prepared by electrolytic reduction (Hg cathode).
Pu^{3+}	Stable to water and air. Oxidizes by action of its own -radiation to Pu^{4+}. Prepared by reduction by SO_2, Zn, U^{4+} or H_2(g) with Pt catalyst.
Md^{3+}	Stable. Can be reduced to Md^{2+}.
No^{3+}	Unstable, reduces to No^{2+}.
Th^{4+}	Stable.
Pa^{4+}	Stable to water. Rapid oxidation by air to Pa(V). Prepared by electrolytic reduction (Hg cathode) and by the action of Zn(Hg), Cr^{2+} or Ti^{3+} in HCl.
U^{4+}	Stable to water. Slow oxidation by air to UO_2^{2+}. Oxidation in nitrate media catalyzed by UV light. Prepared by oxidation of U^{3+} by air, by electrolytic reduction of UO_2^{2+} (Hg cathode) and by reduction of UO_2^{2+} by Zn or H_2(g) with Ni catalyst.
Np^{4+}	Stable to water. Slow oxidation by air to NpO_2^+. Prepared by oxidation of Np^{3+} by air or reduction of higher oxidation states by Fe^{2+}, SO_2, I^- or H_2(g) with Pt catalyst.
Pu^{4+}	Stable in concentrated acids, e.g. 6 M HNO_3. Disproportionates to Pu^{3+} and PuO_2^+ at lower acidities. Prepared by oxidation of Pu^{3+} by BrO_3^- , Ce^{4+}, $Cr_2O_7^{2-}$, HIO_3 or MnO_4^- in acid solution or by reduction of higher oxidation states by HNO_2, NH_3OH^+, I^-, 3 M HI, 3M HNO_3, Fe^{2+}, $C_2O_4^{2-}$ or HCOOH in acid solution.
Am^{4+}	Not stable in water.
Bk^{4+}	Stable to water. Slow reduction to Bk^{3+}. Prepared by oxidation of Bk^{3+} by $Cr_2O_7^{2-}$ or BrO_3^-.
PaO^{3+} or PaO_2^+	Stable. Difficult to reduce.
UO_2^+	Disproportionates to U^{4+} and UO_2^{2+}. Most stable at pH 2.5. Prepared by electrolytic reduction of UO_2^{2+} (Hg cathode) and by reduction of UO_2^{2+} by Zn(Hg) or H_2(g). pH around 2.5 used.
NpO_2^+	Stable. Disproportionates only at high acidities. Prepared by oxidation of lower oxidation states by Cl_2 or ClO_4^- and by reduction of higher oxidation states by NH_2NH_2, NH_2OH, HNO_2, H_2O_2/HNO_3, Sn^{2+} or SO_2.
PuO_2^+	Disproportionates to Pu^{4+} and PuO_2^{2+}. Most stable at low acidities. Prepared by reduction of PuO_2^{2+} by I^- or SO_2 at pH 2.
UO_2^{2+}	Stable. Difficult to reduce.
NpO_2^{2+}	Stable. Easy to reduce. Prepared from lower oxidation states by oxidation by Ce^{4+}, MnO_4^-, Ag^{2+}, Cl_2 or BrO_3^-.
PuO_2^{2+}	Stable. Fairly easy to reduce. Reduces slowly under the action of its own -radiation. Prepared by oxidation of lower oxidation states by BiO_3^-, HOCl or Ag^{2+}.
AmO_2^{2+}	Stable. Reduces fairly rapidly under action of its own α-radiation. Prepared by electrolytic oxidation (Pt anode) in 5 M H_3PO_4, by $S_2O_8^{2-}$ in presence of Ag^+ or by Ag^{2+} in K_2CO_3 solution.

Table 14.5 Characteristic reactions of actinide ions of different valency states with some important anions.[†]

Reagent	Conditions	Precipitated ions	Not precipitated
OH^-	pH ≥ 5	M^{3+}, M^{4+}, MO_2^+, MO_2^{2+}	
F^-	4 M H^+	M^{3+}, M^{4+}	MO_2^+, MO_2^{2+}
IO_3^-	0.1 M H^+	M^{4+} (M^{3+} may oxidize)	MO_2^+, MO_2^{2+}
PO_4^{3-}	0.1 M H^+	M^{4+} (Ac^{3+} partly)	M^{3+} (Pu^{3+} and higher An)
CO_3^{2-}	pH > 10	M^{3+}, M^{4+} (as hydroxide)	MO_2^{2+} (anionic complex)
CH_3COO^-	0.1 M H^+	MO_2^{2+}	M^{3+}, M^{4+}, MO_2^+
$C_2O_4^{2-}$	1 M H^+	M^{3+}, M^{4+}	MO_2^+, MO_2^{2+}

[†]Cl^-, NO_3^-, and SO_4^{2-} do not precipitate actinide ions. An = actinide.

properties have been extensively exploited for the separation and isolation of the individual elements in pure form; see Ch. 21.

The compounds formed are normally quite ionic. The ionic radii of the actinide elements of the different valency states decrease with increasing atomic number (the actinide contraction, Table 14.3). Consequently the charge density of the actinide ions increases with increasing atomic number and, therefore, the probability of formation of complexes and of hydrolysis increases with atomic number. This is illustrated in Figure 14.8, where the heavier actinides are eluted before the lighter ones because the α-hydroxy-isobutyrate eluant forms stronger complexes as the cation radius decreases.

The pattern of stabilities of complexes in the tetravalent states and hexavalent states with hard ligands is the same and follows the order of decreasing ionic radius, Table 14.3.

$$Th^{4+} < U^{4+} < Np^{4+} < Pu^{4+} \tag{14.27}$$

$$UO_2^{2+} < NpO_2^{2+} < PuO_2^{2+} \tag{14.28}$$

This also explains the extraction behavior (cf. §18.4.3) for the M^{4+} (actinide) ions in Figure 14.9, where Pu^{4+} is better extracted than Th^{4+}; the Figure shows the extraction of some actinide ions from HCl or HNO_3 solutions by the different reagents. In the case of tributyl phosphate ,TBP, dissolved in kerosene, the extracted M(VI), M(IV) and M(III) species are $MO_2(NO_3)_2(TBP)_2$, $M(NO_3)_4(TBP)_2$, and $M(NO_3)_3(TBP)_3$, respectively.

For the same element, the stability of the complexes varies with the oxidation state in the series

$$M^{4+} \geq MO_2^{2+} > M^{3+} \geq MO_2^{+} \tag{14.29}$$

The reversal between M^{3+} and MO_2^{2+} reflects that the hexavalent metal atom in the linear $[OMO]^{2+}$ is only partially shielded by the two oxygen atoms; thus the metal ion MO_2^{2+} has a higher charge density than M^{3+} (i.e. about 3.2 ± 0.1). Similarly in MO_2^+, the effective charge is ca. 2.2 ± 0.1. This gives a reasonable explanation of the extraction pattern of Figure 14.9, though other factors (molar volume of the complex etc.)

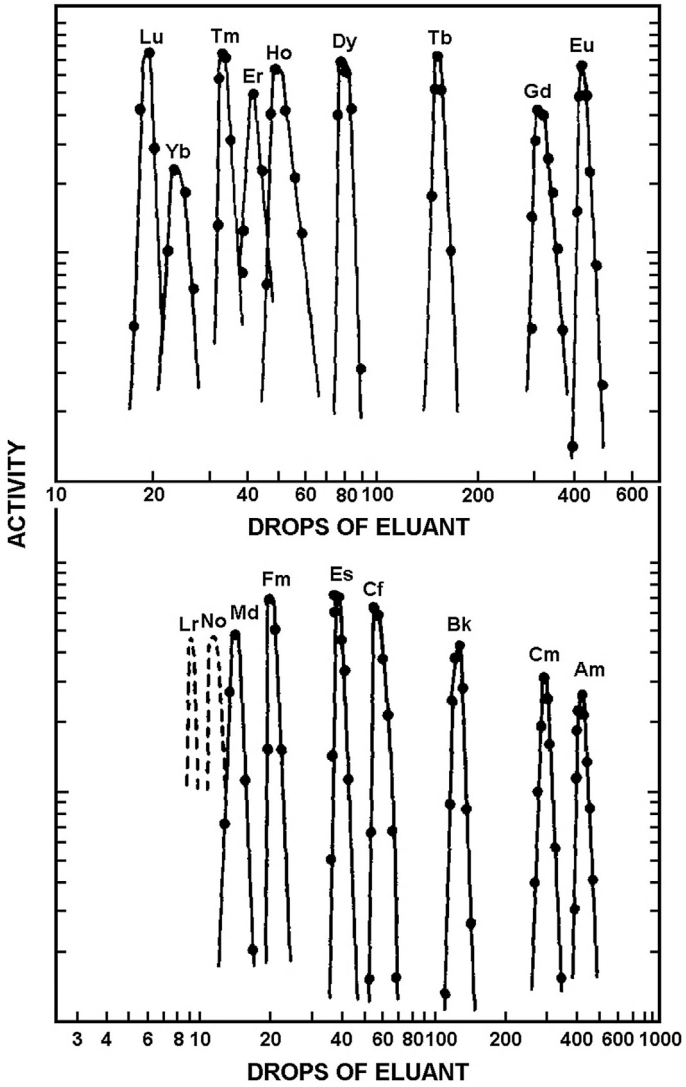


Figure 14.8 Elution curves for 3+ lanthanide and actinide ions from Dowex-50 ion exchange resin with ammonium-α-hydroxy isobutyrate eluant. *(From Katz and Seaborg).*

also contributes. Of importance in reprocessing (§21.6.3) are the low distribution ratios ($D_M \leq 0.01$) of fission products like Cs, Sr, Ru and Zr in the systems in Figure 14.9 (A) and (B).

The extraction of trivalent actinides in general follows the sequence

$$Ac^{3+} < Am^{3+} < Cm^{3+} \text{etc} \tag{14.30}$$

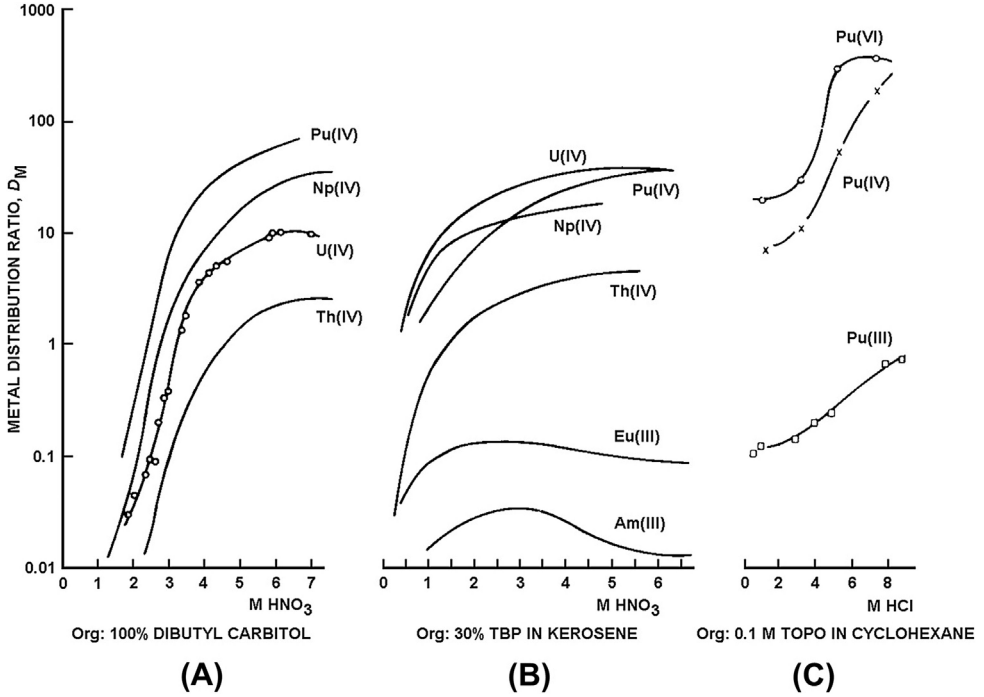


Figure 14.9 Extraction of actinide complexes into various solvents. *(From Ahrland, Liljenzin and Rydberg).*

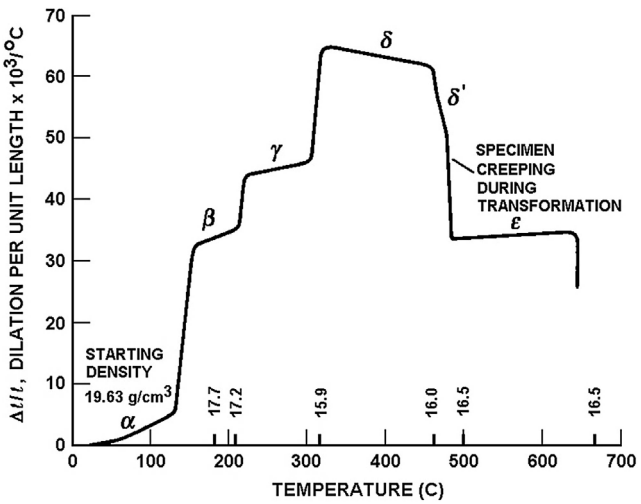


Figure 14.10 Dilation curve and densities of high purity plutonium. *(From Waldron, Garstone, Lee, Mardon, Marples, Poole and Williamson).*

(for example, Cm^{3+} is extracted easier, i.e. at a lower pH, than Am^{3+}, etc.), as the ionic radii decrease in that order.

The chemistry involved in the isolation and purification of the actinide elements from irradiated reactor fuel elements is further discussed in Chapter 21. Actinide chemistry in the ecosphere is discussed in §22.6.

14.4. USES OF ACTINIDES

The use of the actinide elements fall into three categories: (i) for understanding fundamental chemistry and the nature of the periodic system, (ii) as fuels and products in the large scale use of nuclear energy, and (iii) miscellaneous applications, where the particular physical, chemical or nuclear properties are valuable. Only the last aspect is discussed here, the others are treated elsewhere in this book. The availability of transuranium element isotopes suitable for experiments is listed in Table 14.6.

Table 14.6 Availability of transuranium element materials

Nuclide	$t_{½}$	Decay mode	Specific amounts[†] available	Specific activity (Bq/g)
^{237}Np	2.14×10^6 y	α, SF(10^{-10} %)	kg	2.61×10^7
^{238}Pu	87.7 y	α, SF(10^{-7} %)	kg	6.33×10^{11}
^{239}Pu	2.41×10^4 y	α, SF(10^{-4} %)	kg	2.30×10^9
^{240}Pu	6.56×10^3 y	α, SF(10^{-6} %)	kg	8.40×10^{10}
^{241}Pu	14.4 y	β, α(10^{-3} %)	1–10 g	3.82×10^{12}
^{242}Pu	3.76×10^5 y	α, SF(10^{-3} %)	100 g	1.46×10^8
^{244}Pu	8.00×10^7 y	α, SF(10^{-1} %)	10–100 mg	6.52×10^5
^{241}Am	433 y	α , SF(10^{-10} %)	kg	1.27×10^{11}
^{243}Am	7.38×10^3 y	α, SF(10^{-8} %)	10–100 g	7.33×10^9
^{242}Cm	162.9 d	α, SF(10^{-5} %)	100 g	1.23×10^{14}
^{243}Cm	28.5 y	α,ε (0.2 %)	10–100 mg	1.92×10^{12}
^{244}Cm	18.1 y	α, SF(10^{-4} %)	10–100 g	3.00×10^{12}
^{248}Cm	3.40×10^5 y	α, SF(8.3 %)	10–100 mg	1.57×10^8
^{249}Bk	320 d	β,α(10^{-3} %), SF(10^{-8} %)	10–50 mg	6.00×10^{10}
^{249}Cf	350.6 y	α, SF(10^{-7} %)	1–10 mg	1.52×10^{11}
^{250}Cf	13.1 y	α, SF(0.08 %)	10 mg	4.00×10^{12}
^{252}Cf	2.6 y	α, SF(3.1 %)	10–1000 mg	2.00×10^{13}
^{254}Cf	60.5 d	SF, α (0.3 %)	μg	3.17×10^{14}
^{253}Es	20.4 d	α, SF(10^{-5} %)	1–10 mg	9.33×10^{14}
^{254}Es	276 d	α	1–5 μg	6.83×10^{13}
^{257}Fm	100.5 d	α, SF(0.2 %)	1 pg	1.83×10^{14}

[†]According to Oak Ridge Nat. Lab., USA, late 1990:ies.